

# Kejun “Albert” Ying, Ph.D.

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*Studying aging at the intersection of biology and AI*

## Education & Experience

### Stanford University & University of Washington

Stanford, CA & Seattle, WA

*Postdoctoral Researcher*

Spring 2025 – Present

- Co-advised by Dr. Tony Wyss-Coray and Dr. David Baker
- NIH/NIA F99/K00 Fellow

### Harvard University

Cambridge, MA

*Ph.D., Biological Science in Public Health*

Fall 2019 – Spring 2025

- Dissertation: “On the Quantification of Aging”
- Advisor: Dr. Vadim Gladyshev, Harvard Medical School, Brigham and Women’s Hospital

### Harvard University

Cambridge, MA

*M.S., Computational Science Engineering*

Summer 2023 – Spring 2024

- Secondary field during Ph.D. study

### University of California, Berkeley

Berkeley, CA

*Visiting Student, Integrative Biology*

Summer 2017 – Fall 2017

### Sun Yat-Sen University (Yat-Sen Honor School)

Guangzhou, China

*B.S., Life Science*

Fall 2015 – Summer 2019

- Thesis: Screening for the Interactome of hTERC based on Molecular Fluorescence Complementation System in Living Cells
- Advisor: Dr. Zhou Songyang

## Other Professional Experience

### Stanford University & University of Washington

Stanford, CA & Seattle, WA

*Visiting Scholar, Wyss-Coray’s Lab & Baker’s Lab*

Nov 2024 – May 2025

### Harvard Medical School, Boston Children’s Hospital

Boston, MA

*RNA Modifications Graduate Researcher (Rotation), Eric Greer’s Lab*

Jan 2020 – Mar 2020

### Harvard Medical School

Boston, MA

*Cell Reprogramming Graduate Researcher (Rotation), David Sinclair’s Lab*

Oct 2019 – Dec 2019

### Harvard T. H. Chan School of Public Health

Boston, MA

*mTORC1 Graduate Researcher (Rotation), Brendan Manning’s Lab*

July 2019 – Oct 2019

### Undergraduate Research

2015 – 2019

- Sun Yat-Sen University, Telomere & Telomerase
  - University of Edinburgh, Population genetics
  - University of Washington, Acarbose & Rapamycin
  - Buck Institute for Research on Aging, Senolytics
  - University of California, Berkeley, SIRT7
- Zhou Songyang’s Lab  
Xia Shen’s Lab  
Matt Kaeberlein’s Lab  
Judith Campisi’s Lab  
Danica Chen’s Lab

## Grants

|                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |
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| <b>THRIVE: Transforming Health: Reclaiming Intrinsic Vitality for Everyone</b>                                                                                                                                                                                                                                                                                                                                   | ARPA-H PROSPR              |
| <i>Developing PROSPR Intrinsic Capacity score for predicting 20-year health outcomes,</i>                                                                                                                                                                                                                                                                                                                        | 2026 – 2031                |
| <ul style="list-style-type: none"> <li>• USD \$36 million over 5 years (Key Personnel)</li> <li>• Multi-institutional collaboration: Stanford University, Harvard University, Buck Institute for Research on Aging</li> <li>• Integrating multi-omics biomarkers and functional assessments across 19+ longitudinal databases to enable personalized health monitoring and accelerate clinical trials</li> </ul> |                            |
| <b>Aging and Longevity Award</b>                                                                                                                                                                                                                                                                                                                                                                                 | ResearchHub Foundation RFP |
| <i>Funded proposal selected through open peer review,</i>                                                                                                                                                                                                                                                                                                                                                        | 2026 – 2027                |
| <b>OpenAI Researcher Access Program</b>                                                                                                                                                                                                                                                                                                                                                                          | OpenAI                     |
| <i>Research credit grant supporting AI-driven aging biology and protein design,</i>                                                                                                                                                                                                                                                                                                                              | 2026                       |
| <ul style="list-style-type: none"> <li>• USD \$20,000 in API credits</li> </ul>                                                                                                                                                                                                                                                                                                                                  |                            |
| <b>Koo Post-doctoral Transition Award</b>                                                                                                                                                                                                                                                                                                                                                                        | NIH/NIA                    |
| <i>Using causal aging biomarkers and protein design to develop novel anti-aging interventions,</i>                                                                                                                                                                                                                                                                                                               | 2025 – 2028                |
| <b>F99 Transition to Aging Research for Predoctoral Students</b>                                                                                                                                                                                                                                                                                                                                                 | NIH/NIA                    |
| <i>Using causal aging biomarkers and protein design to develop novel anti-aging interventions,</i>                                                                                                                                                                                                                                                                                                               | 2024 – 2025                |
| <ul style="list-style-type: none"> <li>• Award Document Number: FAG088431A (PI)</li> <li>• Received a <i>perfect</i> Impact Score of <b>10</b></li> </ul>                                                                                                                                                                                                                                                        |                            |

## Publications

<sup>†</sup> Corresponding author; <sup>\*</sup> Co-first author; <sup>+</sup> Contributed as consortium author

### SELECTED PUBLICATIONS & PREPRINTS

**Ying, K.**, Paulson, S., Reinhard, J., ..., Gladyshev, V. N. (2026). An Open Competition for Biomarkers of Aging. **Nature Aging (In Press)**. <https://doi.org/10.1101/2024.10.29.620782>

Moqri, M.<sup>\*</sup>, **Ying, K.**<sup>\*</sup>, Poganik, J.<sup>\*</sup>, Herzog, C.<sup>\*</sup>, ..., Marioni, R. E., Lasky-Su, J. A., Snyder, M. P., & Gladyshev, V. N. (2026). Integrative epigenetics and transcriptomics identify aging genes in human blood. **Nature Communications**, 16, 67369. <https://doi.org/10.1038/s41467-025-67369-1>

**Ying, K.**, Paulson, S., Eames, A., Tyshkovskiy, A., ..., Gladyshev, V. N. (2025). *A Unified Framework for Systematic Curation and Evaluation of Aging Biomarkers*. **Nature Aging**. <https://www.nature.com/articles/s43587-025-00987-y>

Wu, X.<sup>\*</sup>, Liu, H.<sup>\*</sup>, **Ying, K.**<sup>\*†</sup> (2025). Biological Age, Aging Clocks, and the Interplay with Lymphoid Neoplasms: Mechanisms and Clinical Frontiers. **Lymphatics**, 3(3), 19. <https://doi.org/10.3390/lymphatics3030019>

**Ying, K.**<sup>†</sup> (2024). Causal inference for epigenetic ageing. **Nature Reviews Genetics**, 1–1. <https://doi.org/10.1038/s41576-024-00799-7>

**Ying, K.**, Castro, J. P., Shindyapina, A. V., ..., Gladyshev, V. N. (2024). Depletion of loss-of-function germline mutations in centenarians reveals longevity genes. **Nature Communications**, 15(1), 5956. <https://doi.org/10.1038/s41467-024-52967-2>

**Ying, K.**, Liu, H., Tarkhov, A. E., ..., Gladyshev, V. N. (2024). Causality-enriched epigenetic age uncouples damage and adaptation. **Nature Aging (February Cover)**, 1–16. <https://doi.org/10.1038/s43587-023-00557-0>

**Ying, K.**, Zhai, R., Pyrkov, T. V., ..., Gladyshev, V. N. (2021). Genetic and phenotypic analysis of the causal relationship between aging and COVID-19. **Communications Medicine**, 1(1), 35. <https://doi.org/10.1038/s43856-021-00033-z>

**Ying, K.**,<sup>\*†</sup> Tyshkovskiy, A., Moldakozhayev, A., ..., Gladyshev, V. N. (2025). Autonomous AI Agents Discover Aging Interventions from Millions of Molecular Profiles. **bioRxiv (Nature under review)**. <https://doi.org/10.1101/2023.02.28.530532>

**Ying, K.**,<sup>†</sup> Song, J., Cui, H., ..., Gladyshev, V. N.<sup>†</sup> (2024). MethylGPT: a foundation model for the DNA methylome. **bioRxiv (Nature Aging 1st Revision)**. <https://doi.org/10.1101/2024.10.30.621013>

**Ying, K.**, Tyshkovskiy, A., Chen, Q., ..., Gladyshev, V. N. (2024). High-dimensional Ageome Representations of Biological Aging across Functional Modules. **bioRxiv (Nature Aging 2nd Revision)**. <https://doi.org/10.1101/2024.09.21.570935>

#### OTHER PUBLICATIONS

Wu, F., Xuan, W., Qi, H., ..., **Ying, K.**, ..., Choi, Y. (2026). Proteo-R1: Reasoning Foundation Models for De Novo Protein Design. **ICML 2026**. <https://arxiv.org/abs/2605.02937>

Zhavoronkov, A., **Ying, K.**, & Wilczok, D. (2026). Peakspan: Defining, Quantifying and Extending the Boundaries of Peak Productive Lifespan. **Aging and Disease**. <https://doi.org/10.14336/AD.2026.0080>

Kordowitzki, P., & **Ying, K.** (2026). The pursuit of understanding human longevity. **npj Aging**, 12(1), 25. <https://doi.org/10.1038/s41514-026-00339-z>

Mavrommatis, C., Belsky, D. W., **Ying, K.**, Moqri, M., Campbell, A., Richmond, A., Gladyshev, V. N., Chandra, T., McCartney, D. L., & Marioni, R. E. (2025). An unbiased comparison of 14 epigenetic clocks in relation to 174 incident disease outcomes. **Nature Communications**, 16, 11164. <https://doi.org/10.1038/s41467-025-66106-y>

Zhang, O., Lin, H., Zhang, X., Wang, X., Wu, Z., Ye, Q., Zhao, W., Wang, J., **Ying, K.**, Kang, Y., Hsieh, C.-Y., Hou, T. (2025). Graph neural networks in modern AI-aided drug discovery. **Chemical Reviews**, 125, 10001–10103. <https://doi.org/10.1021/acs.chemrev.5b00254>

Zhang, O., ..., **Ying, K.**, Huang, Y., Zhao, H., Kang, Y., Pan, P., Wang, J., Guo, D., Zheng, S., Hsieh, C.-Y., & Hou, T. (2025). ECloudGen: leveraging electron clouds as a latent variable to scale up structure-based molecular design. **Nature Computational Science**. <https://doi.org/10.1038/s43588-025-00886-7>

Farinas, A., Rutledge, J., Bot, V. A., Western, D., **Ying, K.**, Lawrence, K. A., Oh, H. S. H., ..., Wyss-Coray, T. (2025). Disruption of the cerebrospinal fluid–plasma protein balance in cognitive impairment and aging. **Nature Medicine**, 1–12. <https://doi.org/10.1038/s41591-025-03831-3>

Rothi, M.H., Sarkar, G.C., Haddad, J.A., Mitchell, W., **Ying, K.**, et al. (2025). The r8S rRNA methyltransferase DIMT-1 regulates lifespan in the germline later in life. **Nature Communications**, 16, 6944. <https://doi.org/10.1038/s41467-025-62323-7>

Grzeccka, A., Iqbal, S., **Ying, K.**, Kordowitzki, P. (2025). Circular RNAs as regulators and biomarkers of mammalian ovarian ageing. **GeroScience**, 1–19. <https://doi.org/10.1007/s11357-025-01798-0>

Jacques, E., Herzog, C., **Ying, K.**, ... Gladyshev, V. N. (2025). Invigorating discovery and clinical translation of aging biomarkers. **Nature Aging**, 1–5. <https://doi.org/10.1038/s43587-025-00838-w>

Goeminne, L. J. E., Vladimirova, A., Eames, A., Tyshkovskiy, A., Argentieri, M. A., **Ying, K.**, Moqri, M., & Gladyshev, V. N. (2025). Plasma protein-based organ-specific aging and mortality models unveil diseases as accelerated aging of organismal systems. **Cell Metabolism**, <https://doi.org/10.1016/j.cmet.2024.10.005>

Gladyshev, V. N., Anderson, B., Barlit, H., ..., **Ying, K.**, Yunes, J., Zhang, B., & Zhavoronkov, A. (2024). Disagreement on foundational principles of biological aging. **PNAS Nexus**, 3(12), pgae499. <https://doi.org/10.1093/pnasnexus/pgae499>

Lyu, YX.<sup>\*</sup>, Fu, Q.<sup>\*</sup>, Wilczok, D.<sup>\*</sup>, **Ying, K.**<sup>\*</sup>, King, A., ..., Bakula, D. (2024). Longevity biotechnology: Bridging AI, biomarkers, geroscience and clinical applications for healthy longevity. **Aging**, 16(1), 1–25. <https://doi.org/10.18632/aging.206135>

**Biomarkers of Aging Consortium**<sup>+</sup>, Herzog, C. M. S., Goeminne, L. J. E., Poganik, J. R., ..., Gladyshev, V. N. (2024). Challenges and recommendations for the translation of biomarkers of aging. **Nature Aging**, 1–12. <https://doi.org/10.1038/s43587-024-00683-3>

Castro, J. P., Shindyapina, A. V., Barbieri, A., **Ying, K.**, ..., Gladyshev, V. N. (2024). Age-associated clonal B cells drive B cell lymphoma in mice. **Nature Aging**, 4(8), 1–15. <https://doi.org/10.1038/s43587-024-00671-7>

Moqri, M., ..., de Sena Brandine, G., **Ying, K.**, Tarkhov, A., ..., Sebastiano, V. (2024). PRC2-AgeIndex as a universal biomarker of aging and rejuvenation. **Nature Communications**, 15(1), 5956. <https://doi.org/10.1038/s41467-024-50098-2>

Tarkhov, A. E., Lindstrom-Vautrin, T., Zhang, S., **Ying, K.**, Moqri, M., ..., Gladyshev, V. N. (2024). Nature of epigenetic aging from a single-cell perspective. **Nature Aging**, 1–17. <https://doi.org/10.1038/s43587-023-00555-2>

Moqri, M., Herzog, C., Poganik, J. R., **Ying, K.**, ... Ferrucci, L. (2024). Validation of biomarkers of aging. **Nature Medicine**, 1–13. <https://doi.org/10.1038/s41591-023-02784-9>

Griffin, P. T., ..., Kerepesi, C., **Ying, K.**, ..., Sinclair, D. A. (2024). TIME-seq reduces time and cost of DNA methylation measurement for epigenetic clock construction. **Nature Aging**, 1–14. <https://doi.org/10.1038/s43587-023-00555-2>

Moqri, M., Herzog, C., Poganik, J. R., **Biomarkers of Aging Consortium**<sup>+</sup>, ... Gladyshev, V. N. (2023). Biomarkers of aging for the identification and evaluation of longevity interventions. **Cell**, 186(18), 3758–3775. <https://doi.org/10.1016/j.cell.2023.08.003>

Liberman, N., Rothi, M. H., Gerashchenko, M. V., Zorbas, C., Boulias, K., MacWhinnie, F. G., **Ying, A. K.**, Flood Taylor, A., ..., Greer, E. L. (2023). 18S rRNA methyltransferases DIMT1 and BUD23 drive intergenerational hormesis. **Molecular Cell**, 83(18), 3268–3282.e7. <https://doi.org/10.1016/j.molcel.2023.08.014>

Bitto, A., Grillo, A. S., Ito, T. K., Stanaway, I. B., Nguyen, B. M. G., **Ying, K.**, ... Kaeberlein, M. (2023). Acarbose suppresses symptoms of mitochondrial disease in a mouse model of Leigh syndrome. **Nature Metabolism**, 5(6), 955–967. <https://doi.org/10.1038/s42255-023-00815-w>

Emmrich, S., Trapp, A., Tolibzoda Zakusilo, F., Straight, M. E., **Ying, A. K.**, Tyshkovskiy, A., ..., Gorbunova, V. (2022). Characterization of naked mole-rat hematopoiesis reveals unique stem and progenitor cell patterns and neotenic traits. **The EMBO Journal**, 41(15), e109694. <https://doi.org/10.15252/embj.2021109694>

Yang, Z., ..., Guo, H., **Ying, K.**, Gustafsson, S., ..., Shen, X. (2022). Genetic Landscape of the ACE2 Coronavirus Receptor. **Circulation**, 145(18), 1398–1411. <https://doi.org/10.1161/CIRCULATIONAHA.121.057888>

Li, T., Ning, Z., Yang, Z., Zhai, R., Zheng, C., Xu, W., Wang, Y., **Ying, K.**, Chen, Y., & Shen, X. (2021). Total genetic contribution assessment across the human genome. **Nature Communications**, 12(1), 2845. <https://doi.org/10.1038/s41467-021-23124-w>

Zhu, J., Xu, M., Liu, Y., Zhuang, L., **Ying, K.**, Liu, F., ..., Songyang, Z. (2019). Phosphorylation of PLIN3 by AMPK promotes dispersion of lipid droplets during starvation. **Protein & Cell**, 10(5), 382–387. <https://doi.org/10.1007/s13238-018-0593-9>

#### OTHER PREPRINTS

Eames, A., Glubokov, D., Moldakozhayev, A., Yücel, A. D., Tyshkovskiy, A., **Ying, K.**, Goeminne, L. J. E., de Magalhães, C. G., & Gladyshev, V. N. (2026). Anatomy of aging through organ-resolved multi-modal imaging and deep learning. **medRxiv**. <https://doi.org/10.64898/2026.03.14.26348392>

Roberts, K. F., Abrams, Z. B., Cappelletti, L., Moqri, M., ..., **Ying, K.**, ..., Reese, J. T. (2026). OpenScientist: evaluating an open agentic AI co-scientist to accelerate biomedical discovery. **medRxiv**. <https://doi.org/10.64898/2026.03.15.26348338>

Wu, Y. C., Yin, M., Shi, B., Zhang, Z., ..., **Ying, K. A.**, ..., Wang, M., & Cong, L. (2026). MedOS: AI-XR-Cobot World Model for Clinical Perception and Action. **medRxiv**. <https://doi.org/10.64898/2026.02.18.26345936>

Zhang, O., Zhang, X., Lin, H., Tan, C., Wang, Q., Mo, Y., ..., **Ying, K.**, Li, J., Zeng, Y., Lang, L., Pan, P., Cao, H., Song, Z., Qiang, B., Wang, J., Ji, P., Bai, L., Zhang, J., Hsieh, C.-Y., Heng, P. A., Sun, S., Hou, T., & Zheng, S. (2025). ODesign: A World Model for Biomolecular Interaction Design. **arXiv**. <https://odesign1.github.io/>

Mavrommatis, C., Belsky, D., **Ying, K.**, Moqri, M., Campbell, A., Richmond, A., ..., Gladyshev, V. N. (2025). *An unbiased comparison of 14 epigenetic clocks in relation to 10-year onset of 174 disease outcomes in 18,859 individuals*. **medRxiv**. <https://doi.org/10.1101/2025.07.14.25331494>

Galkin, F., ..., Tyshkovskiy, A., **Ying, K.**, Gladyshev, V. N., & Zhavoronkov, A. (2024). Precious3GPT: Multimodal Multi-Species Multi-Omics Multi-Tissue Transformer for Aging Research and Drug Discovery. **bioRxiv**. <https://doi.org/10.1101/2024.07.25.605062>

Tyshkovskiy, A., Kholdina, D., **Ying, K.**, Davitadze, M., ..., Gladyshev, V. N. (2024). Transcriptomic Hallmarks of Mortality Reveal Universal and Specific Mechanisms of Aging, Chronic Disease, and Rejuvenation. **bioRxiv**. <https://doi.org/10.1101/2024.07.04.601982>

Rothi, M. H., Sarkar, G. C., Al Haddad, J., Mitchell, W., **Ying, K.**, Pohl, N., ..., Greer, E. L. (2024). The t8S rRNA Methyltransferase DIMT-1 Regulates Lifespan in the Germline Later in Life. **bioRxiv**. <https://doi.org/10.1101/2024.05.15.570935>

Zhang, B., Tarkhov, A. E., Ratzan, W., **Ying, K.**, Moqri, M., ..., Gladyshev, V. N. (2022). *Epigenetic profiling and incidence of disrupted development point to gastrulation as aging ground zero in Xenopus laevis*. **bioRxiv**. <https://doi.org/10.1101/2022.08.02.502559>

#### Patents

V. N. Gladyshev, **K. Ying**, “High-dimensional measurement of biologic age” (2024). *Provisional Patent Application*

V. N. Gladyshev, **K. Ying**, “Mapping CpG sites to quantify aging traits” (2024). *WO2024039905A2*

#### Software and Database

**Autonomous Lab** (2026) <https://github.com/albert-ying/autonomous-lab>  
*MCP server for multi-agent autonomous workflows; AI agents execute senior-junior loops while humans steer decisions.*

**ClockBase Agent** (2025) <https://www.clockbase.org/>  
*AI agent system that autonomously discovers aging-modifying interventions from ~2M human and mouse molecular profiles.*

**MethylGPT** (2024) <https://github.com/albert-ying/MethylGPT>  
*Foundation model for the DNA methylome, pretrained on 226K+ methylation profiles from 5,281 datasets.*

**Biolearn** (2024) <https://bio-learn.github.io/>  
*Open-source Python library for biomarkers of aging analysis with reference implementations of common aging clocks.*

**ClockBase** (2023) <http://gladyshevlab.org:3838/ClockBase/>  
*Comprehensive platform for biological age profiling across multiple clocks in human and mouse (~300K samples).*

## Presentations

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### ORAL PRESENTATIONS

|                                                                                                                                                                     |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Biomarkers of Aging Symposium 2025</b><br><i>Massive AI agent mining of aging-modifying interventions from millions of molecular profile</i>                     | Boston, MA<br>2025          |
| <b>ASHG 2025 Annual Meeting</b><br><i>Decoding the Aging Methylome: From Causal Inference to Foundation Models</i>                                                  | Boston, MA<br>2025          |
| <b>6th TimePie Longevity Forum</b><br><i>Massive AI agent mining of aging-modifying interventions from millions of molecular profile</i>                            | Shanghai, China<br>2025     |
| <b>CSH-Asia Conference: Stem cell, Aging and Rejuvenation</b><br><i>Massive AI agent mining of aging-modifying interventions from millions of molecular profile</i> | Suzhou, China<br>2025       |
| <b>Keystone Symposia: Aging: New Frontiers in Rejuvenation and Gerotherapeutics</b><br><i>MethylGPT: A Foundation Model for the DNA Methylome</i>                   | Breckenridge, CO<br>2025    |
| <b>Biomarkers of Aging Symposium</b><br><i>Standardization of aging biomarkers and BoA challenge</i>                                                                | Boston, MA<br>2024          |
| <b>Harvard GRIP Presentations</b><br><i>Causal Aging Biomarker empowers Unbiased Anti-Aging Therapy Screening</i>                                                   | Boston, MA<br>2024          |
| <b>4th TimePie Longevity Forum</b><br><i>Causal Aging Biomarker as a Tool for Unbiased Anti-Aging Therapy Screening</i>                                             | Shanghai, China<br>2023     |
| <b>Global Congress on Aesthetic and Anti-Aging (GCAA2023)</b><br><i>Causal Aging Biomarker as a Tool for Unbiased Anti-Aging Therapy Screening</i>                  | Singapore<br>2023           |
| <b>10th Aging Research and Drug Discovery conference (ARDD2023)</b><br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                                 | Copenhagen, Denmark<br>2023 |
| <b>AGE 2023 51st Annual Meeting</b><br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                                                                 | Oklahoma City, OK<br>2023   |
| <b>Broad Institute MPG Retreat</b><br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                                                                  | Cambridge, MA<br>2023       |

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| <b>Harvard GRIP Presentations</b><br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                                                      | Boston, MA<br>2022         |
| <b>Targeting Metabesity 2022</b> , ‘Honorable Mention’<br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                                 | Virtual Conference<br>2022 |
| <b>GSA 2021 Annual Scientific Meeting</b><br><i>Genetic and phenotypic evidence for causal relationships between aging and COVID-19</i>                | Virtual Conference<br>2021 |
| INVITED TALKS                                                                                                                                          |                            |
| <b>St. Jude Children’s Research Hospital</b> , hosted by Dr. Zhaoming Wang<br><i>MethylGPT and Causality-enriched Epigenetic Clock</i>                 | Memphis, TN<br>2025        |
| <b>The Alliance for Longevity Initiatives Scientist Spotlight</b> ,<br><i>Episode 14: Albert Ying</i>                                                  | Online Podcast<br>2024     |
| <b>BioAge Seminar</b> , hosted by Dr. Robert Hughes & Dr. Paul Timmers<br><i>Ageome: Biological age with higher-dimensionality</i>                     | Boston, MA<br>2024         |
| <b>MRC Integrative Epidemiology Unit Seminar</b><br><i>Epigenetic Clocks and Mendelian Randomization</i>                                               | Bristol, UK<br>2024        |
| <b>NIA EL Projects Joint Meeting</b> , National Institute on Aging<br><i>Aging Clocks</i>                                                              | Online Webinar<br>2024     |
| <b>Biomarkers of Aging Challenge</b> , Foresight Institute<br><i>Update Webinar with Foresight</i>                                                     | Online Webinar<br>2024     |
| <b>Everything Epigenetics</b> , podcast hosted by Hannah Went<br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                          | Online Podcast<br>2024     |
| <b>Chinese University of Hong Kong</b> , hosted by Dr. Xin Wang<br><i>Causal Aging Biomarker as a Tool for Systemic Anti-Aging Therapy Screening</i>   | Hong Kong, China<br>2024   |
| <b>Everything Epigenetics</b> , podcast hosted by Hannah Went<br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                          | Online Podcast<br>2023     |
| <b>Chinese University of Hong Kong</b> , hosted by Dr. Xin Wang<br><i>Causal Aging Biomarker as a Tool for Systemic Anti-Aging Therapy Screening</i>   | Hong Kong, China<br>2023   |
| <b>Peking University</b> , hosted by Dr. Jingdong Han<br><i>Causal Aging Biomarker and ClockBase</i>                                                   | Beijing, China<br>2023     |
| <b>Chinese Academy of Sciences</b> , hosted by Dr. Xuming Zhou<br><i>Causal Epigenetic Age Uncouples Damage and Adaptation</i>                         | Beijing, China<br>2022     |
| <b>Foresight Institute</b> , hosted by Allison Duettmann<br><i>Genetic Variation, Aging &amp; Relationship to COVID-19   Joris Deelen, Albert Ying</i> | Online Seminar<br>2020     |

## Honors

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|-----------------------------------------------------------------------------------------------------|------|
| <b>Butler-Williams Scholar</b> (Selected but unable to attend), National Institute on Aging (NIH)   | 2026 |
| <b>Semifinalist</b> , Harvard President’s Innovation Challenge, Health Care and Life Sciences Track | 2025 |

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|----------------------------------------------------------------------|-------------|
| <b>Best Poster Award</b> , Inaugural Biomarker of Aging Symposium    | 2023        |
| <b>Best Poster Award</b> , Gordon Research Conference, Systems Aging | 2022        |
| <b>Hackathon Winner</b> , Longevity Hackathon, VitaDAO               | 2021        |
| <b>Yat-Sen Honor School Program</b> , Sun Yat-Sen University         | 2016 – 2019 |
| <b>Yat-Sen Scholarship</b> , Sun Yat-Sen University                  | 2016 – 2019 |

## Service & Leadership

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|-------------------------------------------------------------------------------|----------------|
| <b>Board Member</b> , Lifeboat Foundation                                     | 2025 – Present |
| <b>Member</b> , Norn Longevity Nexus                                          | 2025 – Present |
| <b>Jury and Mentor</b> , Agentic AI Against Aging Hackathon                   | 2025           |
| <b>Core Member</b> , Biomarkers of Aging Consortium                           | 2024 – Present |
| <b>Organizer</b> , Biomarker of Aging Challenge                               | 2024 – Present |
| <b>President</b> , Harvard Interdisciplinary Discussion on Disease and Health | 2024 – 2025    |
| <b>Agenda Contributor</b> , World Economic Forum                              | 2024           |
| <b>Organizing Committee Member</b> , Biomarker of Aging Symposium 2024        | 2024           |
| <b>Organizing Committee Member</b> , Biomarker of Aging Symposium 2023        | 2023           |

## TEACHING & MENTORING

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|--------------------------------------------------------------------------|-------------|
| <b>Mentor</b> , Yuanpei Young Scholars Program                           | 2023 – 2024 |
| <b>Instructor</b> , Harvard Public Health Symposium For Young Generation | 2023        |

## STUDENTS SUPERVISED

**Predoctoral Students:** Ali Doga Yucel, Siyuan Li, Hanna Liu, Donghyun Lee, Yikun Zhang

## JOURNALS REVIEWED

*Nature, Nature Aging, Nature Communications, Communications Medicine, Genome Medicine, BMC Nephrology, Scientific Reports, ...*

## Selected Press & Media

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|----------------------------------------------------------------------------------------------------------------------|----------|
| <b>Science</b> — Eric Topol, “Learning the language of life with AI”<br>Featuring MethylGPT.                         | Jan 2025 |
| <b>Nature Aging</b> — “Epigenetic clock work ticks forward”<br>Expert commentary featuring causal epigenetic clocks. | Feb 2024 |
| <b>Harvard Gazette</b> — “Looking to rewind the aging clock”<br>Featuring causal epigenetic clocks.                  | Feb 2024 |

**World Economic Forum** — “Not just a number: Science is redefining how we understand ageing” Nov 2024  
Featuring aging clocks for global policy audience.

**Nature Biotechnology** — “Scientists hone tools to measure aging and rejuvenation interventions” Aug 2023  
Featuring ClockBase platform for aging intervention screening.

**Lifespan.io** — “An AI-Based System Has Found a Potential Longevity Drug” Dec 2025  
Featuring ClockBase Agent and ouabain discovery.

**MIT Technology Review** — “A 1990s-Born Chinese Scientist Sparks Anti-Aging Breakthrough” Dec 2025  
Featuring AI agent-driven aging intervention discovery.

**PÚBLICO** (Portugal’s leading newspaper) — “O que dizem os genes de quem tem 100 anos” Dec 2024 Featuring centenarian longevity genes.

**Diário de Notícias** (Portugal) — “Que segredos escondem os genes de quem vive até aos 100 anos?” Dec 2024

**ScienceDaily** — “New epigenetic clocks reinvent how we measure age” Feb 2024

## References

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